Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I am making a Study Planner website for my project and I expect to use concepts such as lists, dataclasses, for loops, conditionals, functions, and some drafter to help me complete this project. I want to improve my general skills across all the concepts that I learned so far in this course, like using lists and dataclasses together and working on my skills of fixing errors of code.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the project will be creating the dataclasses and storing the assignments in a list. I think this will be easy because I've had a lot of experience with this so far in this course and it's more familiar to me than other concepts in this course. I think the hardest part will be designing the website by using drafter and taking the user input to make my code for the website. I think this will be hard because it will require me to use many different concepts together and using drafter and user inputs have been something that I haven't used much so far in this course.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think learning to use drafter to make an interactive website will be the hardest and will take the most time and effort. Some features or components that can be implemented first are creating the assignment dataclasses and lists.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions include general distractions like technology and procrastinating on my project. I will try to minimize them by trying to work on short chunks of my project frequently so I make steady progress and don't leave myself rushing towards the end. I will try to refer to previous course material to help me out while working on the project and also sometimes reach out to professors and TA's for help. I don't have many questions yet about the assignment requirements, but I wonder how much of each requirement do I need or can I just have one small portion of each requirement in my project?

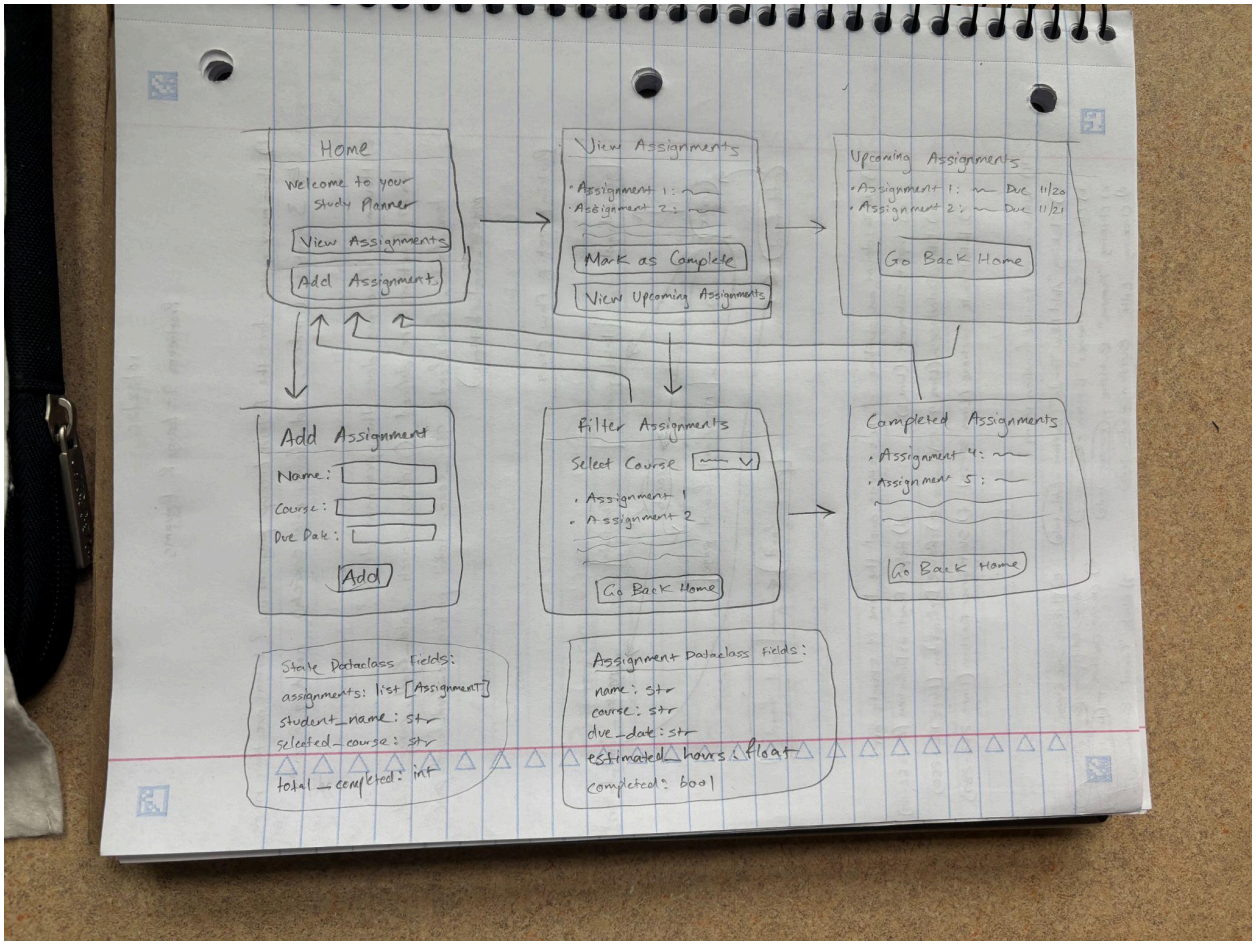
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Home
Welcome to your Study Planner
View Assignments
Add Assignment
View Assignments
• Assignment 1:
• Assignment 2:
Mark as Complete
View Upcoming Assignments
Upcoming Assignments
• Assignment 1: Due 11/20
• Assignment 2: Due 11/21
Go Back Home
Add Assignment
Name:
Course:
Due Date:
Add
Filter Assignments
Select Course
• Assignment 1
• Assignment 2
Go Back Home
Completed Assignments
• Assignment 4:
• Assignment 5:
Go Back Home
State Dataclass Fields:
assignments: list[Assignment]
student_name: str
selected_course: str
total_completed: int
Assignment Dataclass Fields:
name: str
course: str
due_date: str
estimated_hours: float
completed: bool

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of my website is to help students organize their assignments into an organized, easy-to-use website. Students who use the website can add their assignments and due dates, see incomplete and completed work, and see which assignments they should prioritize to complete based on their due dates.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	assignments: list[Assignment] - This will store all the assignments from the user and assignments can be added or removed from the list Student_name: str - This will store the student's name selected_course: str - This will store the course currently selected by the student and can be used when filtering from different courses total_completed: int - This shows the amount of assignments the student has completed
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Assignment dataclass: 1. name: str - name of the assignment 2. course: str - the course of the assignment 3. due_date: str - the due date of the assignment 4. estimated_hours: float - the estimated amount of time (hours) needed to complete the assignment 5. completed: bool - indicates whether the assignment is completed or not

Routes (minimum 7) The names of your route functions	1. home - shows the homepage 2. add_assignment - displays the section of the website to add an assignment 3. submit_assignment - adds an assignment to a list of submitted assignments 4. view_assignments - shows all the assignments in the study planner 5. complete_assignment - allows the user to mark an assignment as completed 6. completed_assignments - shows the assignments that have been completed 7. upcoming_deadlines - shows assignments with approaching due dates 8. filter_assignments - allows the user to see assignments from a specific course
Pages (minimum 5) A brief description of the pages in your site	1. Homepage - the main page of the website 2. Add assignment page - a page where users can add their assignments and the assignment information 3. Upcoming deadlines page - shows a page of assignments with approaching deadlines 4. Completed assignments page - shows a page of assignments that have been completed 5. Filtered course page - shows a page of assignments for a specific course
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	1. TextBox - allows users to enter the assignment name along with other information 2. CheckBox - allows users to check an assignment as completed 3. SelectBox - allows users to select a course's assignments while filtering
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully	1. If an assignment is completed or not - this checks whether or not an assignment is completed 2. If a deadline is approaching soon or not - this

change the logic of the site	checks to see if a certain assignment's due date is approaching soon or not 3. Filtering courses - this checks whether an assignment belongs to the course that was filtered
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	I will use a for loop to go through the assignments in a list and the loop will go through each part of the assignment details and give an output that the user wants.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Me: I am making a study planner website for my computer science project. What are ways I can use a for loop in my website?

LLM: Here are ways you can use for loops in your study planner website:

1. **Display all assignments:** Iterate through the assignment list and display each assignment's name, course, and due date.
2. **Display incomplete assignments:** Go through the list and display only assignments that haven't been completed.
3. **Count completed assignments:** Iterate through the list and count how many assignments the student has finished.
4. **Identify upcoming deadlines:** Go through each assignment and identify which ones are due soon.
5. **Filter assignments by course:** Iterate through the list and display only assignments belonging to a specific course.
6. **Calculate total study hours:** Go through the assignments and add up the estimated hours needed to complete them.

My recommendation: Use ideas 1, 2, and 3 as your main loop implementations. They're simple, practical, and directly related to your website's purpose.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.